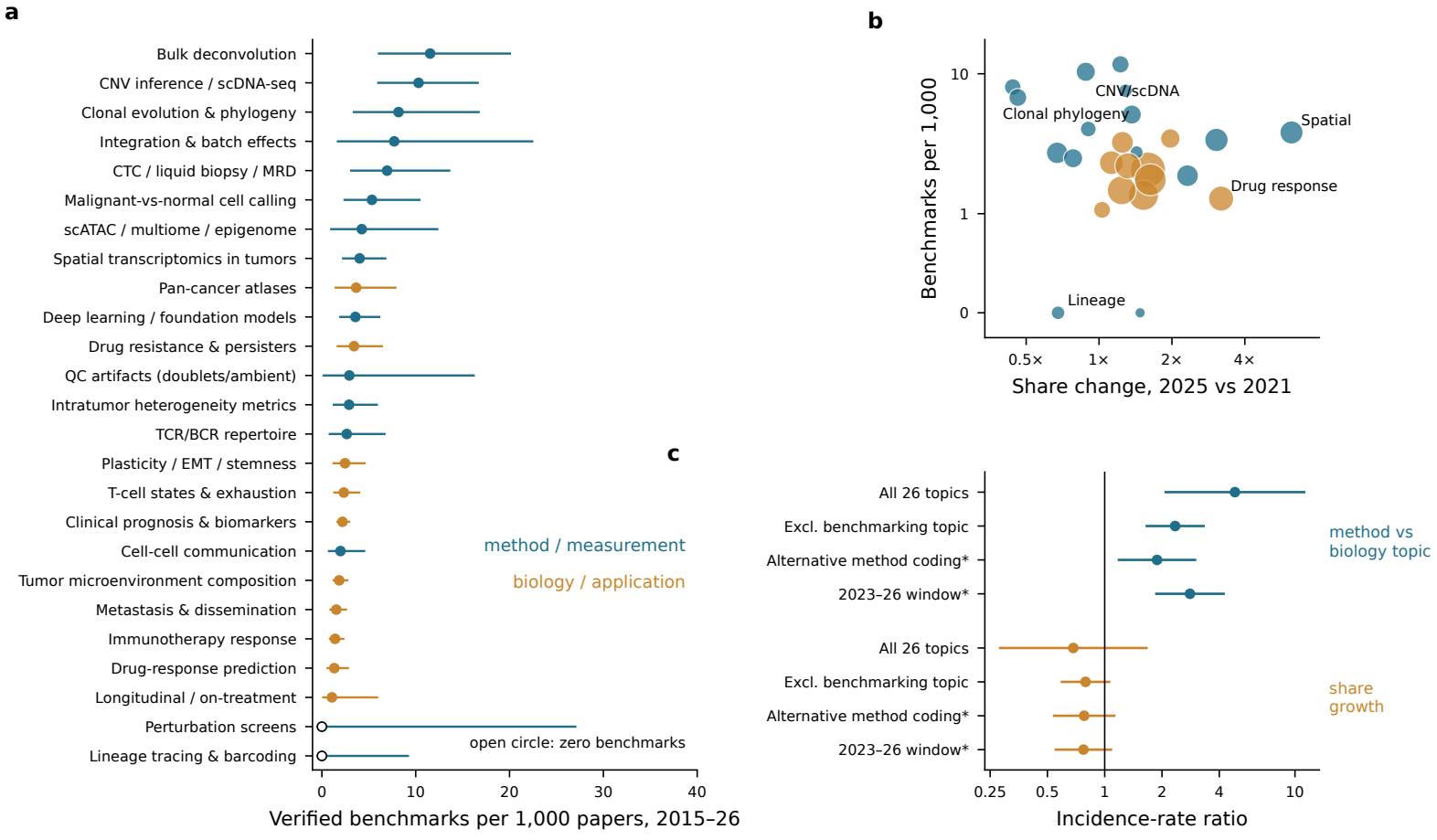
a
Bulk deconvolution
CNV inference / scDNA-seq
Clonal evolution & phylogeny
Integration & batch effects
CTC / liquid biopsy / MRD
Malignant-vs-normal cell calling
scATAC / multiome / epigenome
Spatial transcriptomics in tumors
Pan-cancer atlases
Deep learning / foundation models
Drug resistance & persisters
QC artifacts (doublets/ambient)
Intratumor heterogeneity metrics
TCR/BCR repertoire
Plasticity / EMT / stemness
T-cell states & exhaustion
Clinical prognosis & biomarkers
Cell-cell communication
Tumor microenvironment composition
Metastasis & dissemination
Immunotherapy response
Drug-response prediction
Longitudinal / on-treatment
Perturbation screens
Lineage tracing & barcoding
method / measurement
biology / application
open circle: zero benchmarks
0
10
20
30
40
Verified benchmarks per 1,000 papers, 2015–26
b
Benchmarks per 1,000
10
1
0
CNV/scDNA
Clonal phylogeny
Spatial
Drug response
Lineage
0.5×
1×
2×
4×
Share change, 2025 vs 2021
c
All 26 topics
Excl. benchmarking topic
Alternative method coding*
2023–26 window*
All 26 topics
Excl. benchmarking topic
Alternative method coding*
2023–26 window*
method vs biology topic
share growth
0.25
0.5
1
2
4
10
Incidence-rate ratio